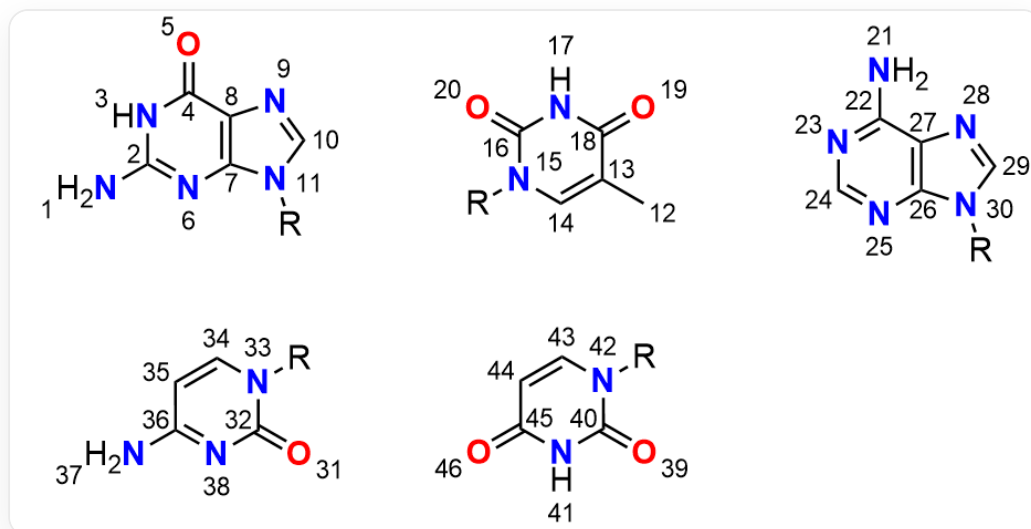


Question

The five nucleotide bases are shown in the figure, where the atomic numbering of heavy atoms is indicated.



The SMARTS of the bases in the figure can be represented as [N:1][C:2]([N:3][C:4]1=[O:5])=[N:6][C:7]2=[C:8]1[N:9]=[C:10][N:11]2[R].[C:12][C:13]3=[C:14][N:15]([R])[C:16]([N:17][C:18]3=[O:19])=[O:20].[N:21][C:22]4=[N:23][C:24]=[N:25][C:26]5=[C:27]4[N:28]=[C:29][N:30]5[R].[O:31]=[C:32]6[N:33]([R])[C:34]=[C:35][C:36]([N:37])=[N:38]6.[O:39]=[C:40]([N:41]7)[N:42]([R])[C:43]=[C:44][C:45]7=[O:46]

Identify and match the above bases. In all base-pairing relationships in DNA/RNA, the base-pairing relationships with fewer hydrogen bonds are denoted as D_1/R_1 , while those with more hydrogen bonds are denoted as D_2/R_2 . For each hydrogen bond in the paired bases $D_1/R_1/D_2/R_2$, the atomic number of the hydrogen bond donor is denoted as h_n ($n = 1 \sim 2/3$), and the atomic number of the hydrogen bond acceptor is denoted as a_n ($n = 1 \sim 2/3$), where n is sorted in ascending order of $a \times h$. For the hydrogen bond pairing in the bases D_1/R_1 , calculate $x_{D_1/R_1} = (a_2 - a_1) \times (h_2 - h_1)$. For the hydrogen bond pairing in the bases D_2/R_2 , calculate $y_{D_2/R_2} = (a_3 - a_1) \times (h_3 - h_2) - a_2 \times h_2$.

Finally, compute the value of $z = \frac{x_{R_1} - y_{D_2}}{x_{D_1} - y_{R_2}}$.

A. 0.55

B. 0.11

C. 1.40

D. -0.14

E. -0.55

F. 0.14

G. -1.23

H. -0.20

Answer

Correct Answer: **A**

Detailed Explanation

The five bases in the figure are, from left to right and top to bottom: guanine (G, atoms 1-11), thymine (T, atoms 12-20), adenine (A, atoms 21-30), cytosine (C, atoms 31-38), and uracil (U, atoms 39-46).

CHECKPOINT

2 PTS

Atoms 1-11 are guanine (G), atoms 12-20 are thymine (T), atoms 21-30 are adenine (A), atoms 31-38 are cytosine (C), and atoms 39-46 are uracil (U)

In DNA, the base pair with two hydrogen bonds, D_1 , is adenine (A, atoms 21-30) and thymine (T, atoms 12-20).

CHECKPOINT

0.5 PTS

D_1 is the pairing of adenine and thymine

The hydrogen bonds between them are $h = 21, a = 19$ and $h = 17, a = 23$. For sorting, we calculate the product $a \times h$: $23 \times 17 = 391$ and $19 \times 21 = 399$. Since 391 is smaller, the first pair is $a_1 = 23, h_1 = 17$, and the second pair is $a_2 = 19, h_2 = 21$.

CHECKPOINT

0.5 PTS

D_1 has $a_1 = 23, h_1 = 17$ and $a_2 = 19, h_2 = 21$

Substituting into the formula $x = (a_2 - a_1) \times (h_2 - h_1)$, we obtain $x_{D_1} = (19 - 23) \times (21 - 17) = -16$.

CHECKPOINT

1 PTS

$x_{D_1} = -16$

In RNA, the base pair with two hydrogen bonds, R_1 , is adenine (A, atoms 21-30) and uracil (U, atoms 39-46).

CHECKPOINT

0.5 PTS

R_1 is the pairing of adenine and uracil

Its hydrogen bonds are $h = 21, a = 46$ and $h = 41, a = 23$. The corresponding products are $23 \times 41 = 943$ and $46 \times 21 = 966$. After sorting, the first pair is $a_1 = 23, h_1 = 41$, and the second pair is $a_2 = 46, h_2 = 21$.

CHECKPOINT

0.5 PTS

R_1 has $a_1 = 23, h_1 = 41$ and $a_2 = 46, h_2 = 21$

Calculation yields $x_{R_1} = (46 - 23) \times (21 - 41) = -460$.

CHECKPOINT

1 PTS

$$x_{R_1} = -460$$

The base pair with three hydrogen bonds, D_2 and R_2 , is guanine (G, atoms 1-11) and cytosine (C, atoms 31-38).

CHECKPOINT

1 PTS

D_2/R_2 is the pairing of guanine and cytosine

This pairing has the same structure in both DNA and RNA, so the values of y_{D_2} and y_{R_2} are identical. The three given hydrogen bonds are $a = 5, h = 37$; $a = 38, h = 3$; and $a = 31, h = 1$. Their $a \times h$ products are $5 \times 37 = 185$, $38 \times 3 = 114$, and $31 \times 1 = 31$. Sorting from smallest to largest, we get: the first pair $a_1 = 31, h_1 = 1$; the second pair $a_2 = 38, h_2 = 3$; and the third pair $a_3 = 5, h_3 = 37$.

CHECKPOINT

1 PTS

D_2/R_2 has $a_1 = 31, h_1 = 1$, $a_2 = 38, h_2 = 3$, and $a_3 = 5, h_3 = 37$

Using the formula $y = (a_3 - a_1) \times (h_3 - h_2) - a_2 \times h_2$, the calculation is $y_{D_2} = (5 - 31) \times (37 - 3) - (38 \times 3) = -884 - 114 = -998$. Thus, the value of y_{R_2} is also -998.

CHECKPOINT

1 PTS

$$y_{D_2} = y_{R_2} = -998$$

Finally, we substitute all calculated values into the final formula $z = \frac{x_{R_1} - y_{D_2}}{x_{D_1} - y_{R_2}}$. The calculation proceeds as

$$z = \frac{-460 - (-998)}{-16 - (-998)} = \frac{-460 + 998}{-16 + 998} = \frac{538}{982} \approx 0.55.$$

CHECKPOINT

1 PTS

$$z = 0.55$$

Therefore, the correct choice is option **A**.